@demotoprod · REGULATED AI · CHECKLIST

Air-gapped AI deployment checklist

We ran a tutorial agent stack (a multi-agent framework, a graph framework, a chat-model client, an embedding model, a tokenizer) with every outside connection refused, the way a deny-all network behaves. Before any task ran it made 37 connection attempts to three outside hosts: 12 to a telemetry server, 24 to a model hub with 70 seconds of retries, one for an encoding file. With the offline switches set, the same code made one attempt. Three checks before you ship behind a fence.

1 Pull the cable in staging before the fence does

Frameworks fetch what they need at the first call, not at install. A demo on a connected laptop never runs the path that fails on day one inside the fence. In our run the embedding load retried the hub five times with backoff and took 69.5 seconds to give up; the telemetry thread tried 12 times in the background.

- Run the whole demo in staging with all egress denied (a deny-all rule, or a resolver that answers nothing).
- Keep the list of every host it tried and every step that failed. That list is the migration plan.
- Repeat it on every dependency upgrade; a new version can add a new call.

2 Stage every file behind the fence with a manifest and hashes

Model weights, tokenizers, encoding files, Python wheels and container images are all fetched lazily by default. One of the three hosts in our run was a tokenizer fetching its encoding file on first use; it has no offline switch, so the only fix is to carry the file in.

- List every artefact the demo downloaded (the failure list from check one tells you).
- Transfer them with a manifest and a hash per file; verify the hashes on the inside.
- Point the libraries at the local cache (the hub's cache directory, the tokenizer's cache directory, a local wheel index).

3 Switch off every phone-home in config, not only at the firewall

A firewall rule makes a call fail; a config switch makes it never happen. The difference is retries, timeouts and threads that block the task. With the telemetry SDK disabled, the framework's telemetry flag set and the hub in offline mode, the same code made one attempt instead of 37 and failed fast where it had to.

- Disable the telemetry SDK for every signal (OTEL_SDK_DISABLED=true) and the framework's own telemetry flag.
- Put the model hub and the model library in offline mode so a cache miss is an error in milliseconds, not a 70-second retry.
- Turn off tracing, auto-update and licence checks in config; keep the firewall as the second line.